

... which is satisfied. By defn, T is a linear transformation. $\square$

1.9 The Matrix of a Linear Transformation

- Whenever a linear transformation T arises geometrically or is described in words, we usually want a "formula" for $T(x)$.
- We shall show that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is actually a matrix transformation $x \mapsto Ax$.

The key to finding A is to observe that T is completely determined by what it does to the columns of the $n \times n$ identity matrix I_n . $\square$ ∇ Imp

Example 1: The columns of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Suppose T is a linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^2$ such that

$$T(e_1) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} \text{ and } T(e_2) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$$

With no additional information, find a formula for the image of an arbitrary x in $\mathbb{R}^2$.